Cool summers are stormy summers: precession reorganizes extratropical cyclone activity through the quiet season

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ABSTRACT: Extratropical cyclones peak in winter, while orbital precession reshapes the seasonal insolation cycle most strongly in summer. Thus storm tracks might be expected to ignore the precession cycle. This is examined with twelve equilibrium simulations using a coupled climate model AWI-ESM2, spanning a full precession cycle at 30° longitude increments under fixed eccentricity and obliquity. Extratropical cyclones are explicitly tracked based on 6-hourly atmospheric pressure fields for both hemispheres. While the annual-mean cyclone response to precession is muted, substantial seasonal variability emerges. In the North Atlantic, the North Pacific, and the Southern Ocean the response concentrates in local summer, where storm activity ranges across the phases by up to 68% of its climatological mean, while winter changes stay below about 10%. We also find summers spent near perihelion feature suppressed storminess, whereas summers near aphelion experience intensified cyclone activity, yielding antiphase midlatitude storm variability between the two hemispheres over the precession cycle. This signal is carried by cyclone number and position, and it follows the pronounced summer response of the Eady growth rate driven primarily by vertical wind shear in all three basins. Precession therefore shapes extratropical storm tracks mostly through their quiet season (i.e., summer). Summer-sensitive proxy archives should record a strong and hemispherically antiphased storminess signal at the precession period that annually integrating archives would miss. Moreover, monthly storm responses are phase-locked to the calendar timing of perihelion, lagged by 1-2 months due to surface thermal inertia. This fingerprint also appears when we apply the same cyclone tracking algorithm to pre-industrial (PI, 0 ka, perihelion falling in winter), Last Interglacial (LIG, 127 ka, perihelion falling in summer) and mid-Holocene (MH, 6 ka, perihelion falling in autumn), with weaker amplitudes matching their weaker eccentricity.

SIGNIFICANCE STATEMENT: Long orbital cycles drive global climate variability, and the strongest of these, precession, reshapes sunlight mainly in summer, while mid-latitude storms happen mainly in winter. This creates an apparent disconnect between precessional forcing and storm track variability. Climate model simulations covering a full precession cycle show that the orbit nevertheless controls storminess by acting through summer. When perihelion falls in a hemisphere's summer, that summer has fewer mid-latitude storms and the storm belt sits farther poleward, and the two hemispheres evolve in antiphase through the precession cycle. Simulations of the Last Interglacial, when perihelion falls in northern summer, reproduce this pattern. This tells researchers where to look for orbital storm signals in geological records, namely in archives sensitive to summer, not to the yearly average.

1. Introduction

Extratropical cyclones govern midlatitude weather systems. They transport the majority of poleward heat and moisture outside the tropics, supply a substantial fraction of midlatitude precipitation, and aggregate into the canonical storm-track pathways of the North Atlantic, North Pacific, and Southern Ocean (Hoskins and Valdes 1990; Chang et al. 2002; Hoskins and Hodges 2002, 2005). These cyclones develop via baroclinic instability driven by the meridional temperature gradient (Eady 1949), a gradient that reaches its maximum steepness during winter. Cyclone activities consequently peak in the cold season, with consistent wintertime maxima (e.g., frequency) evident in both the Northern Hemispheres (Hoskins and Hodges 2002) and the Southern Hemisphere (Simmonds and Keay 2000). Midlatitude cyclogenesis is therefore most favorable during winter.

Precession, by contrast, exerts its primary forcing on warm seasons. Slow variations in the longitude of perihelion represent the leading orbital control on seasonal insolation cycles. It redistributes incoming solar radiation, and the amplitude of insolation anomalies scales with a season's background solar irradiance (Berger 1978; Laskar et al. 2004). Midlatitude summertime insolation fluctuates by over 100 W m^{-2} across a full precession cycle, whereas midwinter insolation experiences only minor orbital modulation. This stark contrast explains why orbital climate frameworks single out summertime solar forcing as the most impactful component of climate changes (Huybers 2006). Since extratropical cyclone activity is inherently winter-favored, this invites a simple expectation that precession should exert negligible control over midlatitude storm tracks. But this expectation has never been put to a clean test.

Previous studies have looked closely at precessional signals, and they find seasonal shifts in climate. This is well established in monsoon research, from the classic orbital experiments of Kutzbach and Guetter (1986) to single-forcing designs that isolate precession explicitly (Bosmans et al. 2015; Merlis et al. 2013), and cave records show the northern and southern monsoons swinging in antiphase as perihelion migrates through the year (Wang et al. 2008). The extratropics have received far less attention. Kaspar et al. (2007) simulated a stronger and poleward-shifted North Atlantic winter storm track for the Eemian from two orbital time slices. Brayshaw et al. (2010) linked Holocene orbital shifts to North Atlantic and Mediterranean winter storm tracks via changes in baroclinicity. And Varma et al. (2012) found the Southern Hemisphere westerlies migrating under Holocene orbital forcing. Explicit cyclone tracking has entered paleoclimate work mainly at the Last Glacial Maximum, where ice sheets dominate the forcing (Pinto and Ludwig 2020; Raible et al. 2021), and in transient simulations of the last millennium (Hess and Chemke 2025). For tropical cyclones the precessional response has recently been isolated in high-resolution paleoclimate simulations, where it is governed by atmospheric moisture (Raavi et al. 2023). Here our study provides the extratropical counterpart, where cyclone growth is baroclinic. To our knowledge, no previous study has applied explicit cyclone tracking across a full precession cycle to isolate the orbital response of the extratropical storm tracks, and none has asked which season carries the response.

This missing piece of research matters beyond just storm dynamics. Many paleoclimate archives that record past storm activity such as wind-blown dust and sediment, or rainfall linked to westerly winds only capture signals from specific seasons instead of annual averages. If a proxy record mainly reflects one single season, it can either hide or falsely create an orbital signal, depending on which time of year carries the strongest storm response (Laepple et al. 2011). Fully understanding which season holds the clearest precessional storm signal is therefore essential for properly interpreting these geological records. Moreover, different metrics for cyclone behavior, e.g., total storm counts, how much rain each storm produces, and storm intensity, react differently to orbital forcing. Researchers must evaluate these quantities separately before drawing conclusions from any single geological proxy.

To fill this gap, we run 12 equilibrium climate simulations covering a full precession cycle. Every extratropical cyclone in both hemispheres is tracked individually using 6-hour sea level pressure. Our experimental design isolates precession at amplified eccentricity with all other boundary conditions fixed, so that the seasonal storm response to precession forcing alone can be examined. The results are then tested in simulationso of real orbital states, such as PI, MH, and LIG. Section 2 describes our model and simulation, and the method we use for storm tracking and analysis. Section 3 presents the results. We discuss and conclude in section 4 and 5 respectively.

2. Methodology

a. Model and experiments

The simulations are performed with the coupled Earth system model AWI-ESM2 (Sidorenko et al. 2019). The atmosphere component is ECHAM6 at T63 resolution with 47 vertical levels (Stevens et al. 2013), which includes the land surface module JSBACH (Raddatz et al. 2007). The ocean and sea ice component is FESOM2, a finite-volume model formulated on an unstructured mesh (Danilov et al. 2017), with a spatially varying resolution from 25 km over high latitudes and coastal regions, to 150 km over open oceans.

12 equilibrium simulations are analyzed, taken from the idealized precession ensemble introduced by Shi et al. (2025). The longitude of perihelion ω is prescribed at values from 0° to 330° in steps of 30° (labeled as p000, p030..., and p330), while eccentricity is fixed at 0.058 and obliquity at 24.5°. The eccentricity value sits near the upper bound reached during the Quaternary and amplifies the precessional modulation of insolation. All remaining boundary conditions follow the pre-industrial setup, and each phase is branched from a long pre-industrial control. 30 years of 6-hourly output from the equilibrated part of each simulation form the basis of the cyclone analysis. The calendar month of perihelion for each phase is computed from orbital geometry (Berger 1978), and the same formulary provides the insolation fields shown in Fig. 1.

To test our results under realistic orbital configurations, we additionally conducted three equilibrium simulations including a PI (0 ka) control run, a MH (6 ka) experiment, and a LIG (127 ka) experiment. Boundary conditions follow the protocols of PMIP4 (Otto-Bliesner et al. 2017). Orbital parameters are calculated following Berger (1977), and the greenhouse gas concentrations are taken from multi-archive reconstructions from the Greenland and the Antarctic ice cores (Flückiger et al. 2002; Monnin et al. 2004; Schneider et al. 2013; Schilt et al. 2010; Buiron et al. 2011). Each simulation is integrated for 1,000 model years with the trend of simulated global mean surface temperature in the final 100 model years not exceeding ±0.05 K/century. We preserved 6-hourly sea level pressure from each simulation. At 127 ka, perihelion occurs close to the June solstice with an eccentricity of 0.039378; at 6 ka, perihelion falls within September, while the pre-industrial control features perihelion in early January.

b. Cyclone detection

Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. (2021), and Fig. 2 summarizes the procedure with one 6-hourly field of the ensemble. 6-hourly sea level pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two hemispheres are processed separately. A grid cell qualifies as a cyclone center when its pressure p_c lies below that of all 8 neighbors and the surrounding field keeps a sufficient inward gradient. The gradient test compares p_c with the mean pressure $\overline{p}(d)$ on a one-cell ring at distance d from the candidate,

$$\frac{\overline{p}(d) - p_c}{d} \geq \gamma \tag{1}$$

where $d = 1000$ km and $\gamma = 750$ Pa$\,(1000\text{ km})^{-1}$, and candidates over terrain above 1500 m are discarded. Centers closer than 1000 km are treated as one multicenter system. The cyclone area A is obtained by growing closed isobars around each center at a 200-Pa interval until a contour fails to close, and the last closed isobar defines the edge pressure p_e (Fig. 2b). Each cyclone carries the depth and the area-equivalent radius

$$D = p_e - p_c \tag{2}$$

$$R = \sqrt{A/\pi} \tag{3}$$

Tracking links the centers of consecutive 6-hourly fields (Fig. 2c). For every active cyclone a first-guess position is projected from its previous displacement, damped to three quarters,

$$\mathbf{x}_q = \mathbf{x}_t + 0.75\,(\mathbf{x}_t - \mathbf{x}_{t-1}) \tag{4}$$

and a center of the next field qualifies as the continuation when it lies within $v_{\max}\Delta t$ of both the current position $\mathbf{x}_t$ and the projection $\mathbf{x}_q$, with $v_{\max} = 150$ km h^{-1} and $\Delta t = 6$ h, a searching radius of 900 km per step. Among the qualifying candidates the one nearest to the projection continues the track. Centers left unmatched at the new time start tracks as genesis events, and tracks left unmatched end as lysis. Only systems poleward of 30° latitude enter the analysis.

Several track-based diagnostics are derived. Track density counts every 6-hourly cyclone position, system density counts each cyclone once at its mean position, and genesis density counts the first point of each track. Tracks that merely cross the boundaries of the monthly processing windows would appear as spurious genesis or lysis events, and such truncated events are removed. For every cyclone the lifespan, the maximum depth relative to the outermost closed isobar, the minimum central pressure, the mean cyclone area, and the propagation speed are recorded. The same pipeline also yields a storm-day field, the number of days per month during which a grid point lies inside the footprint of a detected cyclone. This pointwise measure of influence time is convenient for maps of the storm distribution and of its displacement. It scales with the product of cyclone number and cyclone area, so frequency statements in this paper rest on track density.

c. Baroclinicity diagnostics

The maximum Eady growth rate serves as the measure of baroclinicity (Eady 1949; Lindzen and Farrell 1980; Hoskins and Valdes 1990). It is computed as $\sigma = 0.31\,|f|\,|\partial U/\partial z|\,N^{-1}$ in the 925–700 hPa layer from monthly wind, temperature, and geopotential height. The response to precession is quantified between the end members p090 and p270, which place perihelion at the June and December solstices. Because the difference fields are dipolar, plain box averages cancel, and the response amplitude is instead defined as the root-mean-square of the difference over a basin, normalized by the climatological growth rate of the season. A substitution decomposition in the style of Camargo et al. (2007) splits the difference field into a vertical-shear factor and a static-stability factor, and the share of each factor is obtained by projecting it onto the full difference pattern. The basin averages use the storm-track boxes defined in the next sub-section.

d. Basins and statistics

Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means are area weighted. The size of the precessional response is reported as the range across the twelve phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each phase, and the significance of the end-member difference maps is assessed with a two-sided Welch t test (Welch 1947) at each grid point.

3. Results

a. Summer sensitivity

The 12 precession-phase annual midlatitude storm climatologies exhibit only minor differences, and phase anomalies relative to the multi-phase ensemble mean manifest as weak, slowly evolving dipoles aligned along major storm track pathways (Fig. 3). The basin-averaged time series quantify the muted amplitude of this annual-mean signal, with across-phase ranges of 10%, 17% and 4% in the North Atlantic, the North Pacific and the Southern Ocean, varying smoothly with ω. We also observe an interhemispheric antiphase.

There is also a pronounced seasonal cycle. Extratropical cyclone activities peak in DJF in Northern Hemisphere basins and JJA across the Southern Ocean, with this seasonal timing unchanged across all orbital setups (Fig. 4a–c; Table 1). Cold-season cyclones also carry greater intensity, with winter peak cyclone depths exceeding summertime values by 70-100 % within the Northern Hemisphere storm basins (not shown). Overall, winter dominates midlatitude cyclone activity uniformly across hemispheres and all orbital phases.

Precession exerts negligible control over annual-mean storm conditions but profoundly reshapes seasonal cyclone variability, and its strongest modulation targets the low-activity warm season. Though each precession phase retains the same wintertime peak in basin-averaged storm track density, cross-phase deviations emerge almost entirely within each basin's local summer (Fig. 4a–c). Seasonal range statistics quantify this stark seasonal contrast in orbital sensitivity (Fig. 4d–f). For the North Pacific, the across-phase range of track density accounts for just 7 % of the phase mean on an annual basis, yet rises to 44 % in boreal summer (JJA). The North Atlantic follows an identical pattern, with annual variability of 7 % versus a summertime range of 25 %. The Southern Ocean mirrors this behavior but shifts the sensitive season to its local summer (DJF), with an annual range of only 4 % compared to a 22 % summer signal. All basins uniformly identify local summer as the most orbital sensitive season, and the small annual signal is what survives of a substantial summer swing after the stable winter is averaged in.

This summertime storm response is locked to orbital geometry. When basin-averaged track-density anomalies are plotted against both calendar month and precession angle ω, the band of peak anomalies traces a diagonal ridge aligned with the calendar timing of perihelion across the full phase space (Fig. 4g–i), with a delay of 1-2 months. Summers spent near perihelion exhibit suppressed storm activity, whereas aphelion summers feature intensified storminess. A single precessional forcing thus drives opposing midlatitude storm responses between the two hemispheres across the orbital cycle, as northern summer perihelion aligns with southern summer aphelion under orbital precession, and vice versa, reversing the insolation control on warm-season storminess between hemispheres.

b. Mechanisms

The pronounced seasonal dependence of orbital storm signals originates directly from insolation forcing. Averaged across the Northern Hemisphere storm-track band (40–70°N), precessional insolation variability peaks at 105 W m^{-2} during boreal summer (JJA) and falls to merely 14 W m^{-2} in boreal winter (DJF). The Southern Hemisphere storm band (40–70°S) exhibits mirror-symmetric behavior, with a maximum seasonal forcing amplitude of 119 W m^{-2} in austral winter (DJF) versus just 47 W m^{-2} in austral summer (JJA) (Fig. 1). The underlying mechanism lies in the multiplicative nature of precessional insolation modulation. Shifting perihelion across the annual cycle alters the Sun–Earth distance for each month, which rescales monthly incoming solar flux by a factor proportional to eccentricity e (approximately $4e$, equaling 23 % for $e = 0.058$). Midwinter at midlatitudes features minimal climatological insolation, leaving little baseline flux to be modulated by precessional shifts. Precession forcing therefore exerts its strongest modulation on local summer in each hemisphere, and barely perturbs deep winter conditions.

Orbital insolation anomalies are translated into extratropical storm activity via shifts in midlatitude baroclinicity. Differences in maximum Eady growth rate between the extreme precession phases $\omega = 90°$ and $\omega = 270°$ reach their peak magnitude within each basin's local summer (Fig. 5). Normalized against seasonal climatological means, summertime Eady growth rate responses hit 31 % in the North Atlantic, a pronounced 92 % in the North Pacific, and 21 % in the Southern Ocean. A decomposition analysis further reveals that vertical wind shear dominates the precessional Eady growth rate anomalies, while static stability contributes negligible signal strength. This is consistent across all storm tracks. Perihelion summers warm the summer hemisphere, flatten the meridional temperature gradient, weaken the thermal-wind shear, and suppress the baroclinic growth that sustains warm-season extratropical cyclones. The storm anomalies appear somewhat poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic waves (Chang et al. 2002).

c. Meridional shift

Spatial difference maps between the two orbital end-members (p090 minus p270) yields weak responses for the annual means (Fig. 6a). By contrast, boreal summer (JJA) exhibits a prominent dipole pattern across Northern Hemisphere basins, with fewer storm days through the midlatitude core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined to an overall summer weakening. In DJF the signal migrates to the Southern Ocean and forms a circumpolar banded structure of opposite sign, marking enhanced midlatitude storm activity and an equatorward shift of the southern storm belt.

The latitudinal displacement of storm tracks driving these dipole anomalies is quantified in Fig. 7. For the North Pacific boreal summer, core storm latitude shifts from 53°N when perihelion falls in northern winter to 61°N when it falls in northern summer, yielding an 8° total poleward migration that peaks precisely at $\omega = 90°$. The North Atlantic summertime storm track undergoes a 4.4° poleward shift following identical orbital behavior, while the Southern Ocean mirrors this response with a 3.8° poleward displacement when perihelion coincides with austral summer (near p270). A perihelion summer therefore weakens storm activity and shifts the storm population poleward, consistent with the poleward shift of baroclinic activity evident in the Eady growth rate dipole (Fig. 5a).

By contrast, the annual-mean track latitude varies by no more than 1.3° in any basin. Winter storm-track latitudes only fluctuate by 1.5–2.8° across the full precession cycle, and their latitudinal extrema do not align with solstitial orbital phases. This further confirms our earlier conclusion that midlatitude winter storms are orbitally insensitive. Spring and autumn display intermediate latitudinal variability with a coherent phase dependence. For example, autumn poleward displacement maximizes near $\omega = 150°$, the orbital configuration that places perihelion in late summer. Latitudinal poleward shifts therefore track the calendar timing of perihelion identically to Fig. 4g–i, with a uniform lag of 1–2 months, consistent with surface thermal inertia. Figure 8 quantifies this basin-dependent lag. All basin-scale storm extrema cluster along basin-specific dashed lines offset from the perihelion reference. The response lags the perihelion calendar by distinct intervals across regions, near 1 month in the North Pacific, 1-2 months in the North Atlantic, and slightly over 2 months in the Southern Ocean.

d. Cyclone precipitation

The hydrological signature of precessional forcing evolves in antiphase with storm dynamical responses (Fig. 9). Detected extratropical cyclones contribute roughly one-third of total midlatitude precipitation across our simulations, consistent with storm precipitation partitioning derived from observational reanalysis datasets (Hawcroft et al. 2012). During boreal summer over the North Pacific, mean precipitation per individual storm rises by as much as 15 % at the orbital phase where total storm frequency reaches its minimum, yielding nearly perfectly opposing variability between the two metrics. Perihelion summers feature warmer, more humid midlatitude atmospheres, which amplifies precipitation intensity within the remaining cyclones; nevertheless, the sharp reduction in storm number dominates the total signal, such that basin-wide summertime precipitation drops by 17 % between the two extreme orbital phases. The fractional share of precipitation originating from cyclones also declines markedly, with a full-cycle range of 9.5 % in North Pacific summer compared to just 3 % for the annual mean. Precessional forcing thus exerts two counteracting controls on extratropical hydroclimate simultaneously. It reduces total cyclone counts while boosting precipitation intensity per storm.

e. Realistic paleosimulations (MH and LIG)

Our results based on the 12 idealized simulations can be validated against the realistic orbital-state simulations (PI, MH, and LIG) introduced in Section 2. In LIG, perihelion occurs near June solstice, and the tracked cyclone responses fully reproduce our core theoretical pattern. Relative to PI, summer storm track density decreases by 8% over the North Atlantic and 11% over the North Pacific; winter cyclone changes remain within approximately 3%. Meanwhile, austral summer storm track density across the Southern Ocean rises by 8%. Northern Hemisphere summer storm tracks shift poleward by 1–2 degrees, while the Southern Hemisphere summer storm belt shifts equatorward (Fig. 10). The summer difference composites show identical weakening and poleward displacement of Northern Hemisphere storm tracks as seen in our amplified idealized end-member simulations (Fig. 11). In MH, perihelion happens in September, leaving seasonal forcing dominated by warm September conditions with cool March conditions. Consistent with this seasonal signal, MH storm variability features fewer, poleward-shifted cyclones in autumn and the opposite trend in spring. The magnitude of storm responses in these paleosimulations is much weaker than that of our amplified end-member contrasts (p090 vs. p270, Fig. 6), consistent with their smaller eccentricity values of the real orbits. Notably, these experiments combine realistic precession with their own obliquity and greenhouse gases, so they test our theoretical prediction under multi-factor natural orbital forcing rather than isolating precessional effects.

4. Discussion

The seasonal preference seen in storm responses originates purely from orbital geometry, independent of model physics. Precession redistributes insolation in proportion to the insolation a season already has, meaning winter hemispheres have very little background insolation for orbital changes to modify (Berger 1978). Fig. 1 clearly illustrates this stark seasonal contrast in forcing strength. Winter insolation varies by only roughly 20 W m^{-2} across the precession cycle, while summer values exceed 125 W m^{-2} within mid-latitude storm-track latitudes. This is model independent and will hold in any climate models as it simply reflects the orbital geometry behind the well-known summer-dominated orbital forcing framework (Huybers 2006). Existing monsoon studies has long confirmed that summer bears the strongest precessional signal (Kutzbach and Guetter 1986; Bosmans et al. 2015). But the new insight of our work lies elsewhere. This summer-focused orbital forcing acts on a winter-dominated weather system, reshaping storm activity exclusively during its normally quiet warm season. Note that the term ”quiet” here refers to the weaker overall intensity of the summer storm track, not its population. Both hemisphere still hosts abundant storm systems in local summer, and they are merely weaker, shallower and move more slowly compared to winter cyclones.

The mechanism that converts summer insolation into summer storms is the thermal wind. When perihelion falls in local summer, the corresponding hemisphere experiences a warming over high latitudes. This flattens the meridional temperature gradient and reduces vertical wind shear that drive baroclinic growth. Our factor decomposition analysis confirms that the shear term carries essentially the whole Eady growth rate response in all three ocean basins, while static stability plays a negligible role. We see similar physical mechanism operating in the modern Northern Hemisphere, where greenhouse warming takes the place of perihelion-driven heating. Stronger summer warming at high latitudes has weakened summertime large-scale circulation, eddy kinetic energy, and the number of strong summer cyclones (Coumou et al. 2015; Chang et al. 2016; Gertler and O'Gorman 2019). Future climate projections show the same shear-based decline of Northern Hemisphere summer cyclone activity and baroclinic energy as the globe warms (Lehmann et al. 2014; Priestley and Catto 2022). But a difference should be noted here, as greenhouse warming is a transient forcing that both hemispheres feel in phase, whereas our simulations are equilibrated and precession swings both hemispheres in antiphase. Even so, both forcings follow one unified rule that is the core theme of this study. Cooler summers host more storm activity.

Compared to summer, storm systems in winter are much more stable. Across all three ocean basins, winter storm activity changes by less than 10% over a full precession cycle, even though winter Eady growth rates shift by 16–18% between the two orbital end-members, and weak winter insolation forcing cannot fully explain this stability. The winter storm track appears saturated in the sense that ample baroclinicity is available in every phase, so that moderate changes in the growth rate do not translate into changes in occurrence. A transient simulation of the last millennium likewise found cyclone counts nearly insensitive to external forcing (Raible et al. 2018; Hess and Chemke 2025), Our findings expand this conclusion to the far larger swings in solar radiation driven by precession, and this insensitivity only holds for wintertime conditions.

Geological records sensitive to summertime conditions within midlatitude storm belts should preserve prominent precession-scale storm signals, whereas proxies that average climate signals across winter or the full year will show only minor orbital imprint. Just as observed for temperature reconstructions, the seasonal bias of a given archive can either generate artificial orbital storm signals or fully obscure true precessional variations (Laepple et al. 2011). Furthermore, summer-sensitive storm records from both hemispheres will fluctuate oppositely over precessional cycles, forming an extratropical equivalent of the well-documented monsoon seesaw (Wang et al. 2008; Erb et al. 2015). Current published sediment records already support the feasibility of testing this framework. Distinct precession-period signals have been detected in Northern Hemisphere midlatitude westerly proxies (Li et al. 2019), as well as North Pacific dust records. Precessional variation appears specifically in moisture-sensitive dust fractions, while the annually averaged dust flux responds primarily to obliquity forcing (Zhong et al. 2024). Meanwhile, dust intensity archives that capture winter-only signals display very weak precessional power (Chen et al. 2014). Regarding the Southern Hemisphere, subtropical margins of the southern westerly belt exhibit clear precession cycles (Lamy et al. 2019), consistent with our results. And the presence or absence of precession signals at individual sites depends on their latitudinal position within the wind belt (Kaiser et al. 2024). Records of the southern westerlies also show orbital-band migrations of the belt itself (Lamy et al. 2010; Varma et al. 2012), and glacial-state cyclone modeling has shown how strongly reconstructed storminess depends on which cyclone property a proxy senses (Pinto and Ludwig 2020).

Although the three storm basins follow the same physical mechanism, they exhibit different regional behaviors, which can be distinguished through separate diagnostic metrics. In local summer, storm frequency and track position vary substantially by 16–68% across the precession cycle, while storm propagation speed remains largely unchanged. By contrast, storm intensity exhibits clear basin-dependent differences. The Southern Ocean shows a 22% swing in storm frequency but only a 4% change in storm intensity. The North Pacific features a systematic reduction in overall storm activity during perihelion summers. This is consistent with previous findings from anthropogenic warming scenarios that storm frequency and intensity respond to external forcing through independent physical mechanism (Bengtsson et al. 2009; Ulbrich et al. 2009; Catto et al. 2019). Hydrological responses further amplify this contrast. Individual storms produce stronger precipitation in summers with reduced cyclone numbers, indicating that hydrological adjustments exceed dynamical changes under orbital forcing, a pattern analogous to that identified under greenhouse warming (Bengtsson et al. 2009). Given that extratropical precipitation is predominantly generated by cyclones and frontal systems (Hawcroft et al. 2012; Catto and Pfahl

2013), this competing effect substantially alters precipitation partitioning. So a precipitation proxy tend to record combined signal of reduced storm occurrence and intensified single-storm rainfall during perihelion summers.

Our study is based on idealized simulations designed to isolate pure precessional forcing. The experimental setup holds eccentricity, obliquity, ice sheets, and greenhouse gas concentrations fixed and does not incorporate transient climate evolution or orbital cross-interactions. As such, our results represent dynamical responses to idealized orbital conditions rather than reconstructions of specific geological epochs. Future work could incorporate time-varying boundary conditions and full orbital combinations to further generalize the storm seasonal response across broader paleoclimate backgrounds.

5. Conclusions

In our study, we carry out a set of equilibrium climate simulations covering a full precession cycle, with explicit tracking of all extratropical cyclones across both hemispheres to isolate precessional forcing. We find that precession regulates extratropical cyclone primarily through local summer, the season with naturally low storm activity, while winter storm tracks has only minor change across all precession phases. Storm activity calms down when perihelion falls in local summer and intensifies when aphelion coincides with local summer. And, as perihelion summer in one hemisphere matches aphelion summer in the other, mid-latitude storminess of the two hemispheres varies in antiphase during a precession cycle. This orbital signal propagates through shear-driven baroclinicity and expresses itself in cyclones numbers and position, both of which are phase locked to the calendar month of perihelion, with a delay of 1-2 months. Our result also bring clear implications for paleoclimate proxy interpretation. Orbital storm signals should be extracted from midlatitude archives sensitive to summer climate, and the recorded signals will display opposite variations between the two hemispheres.

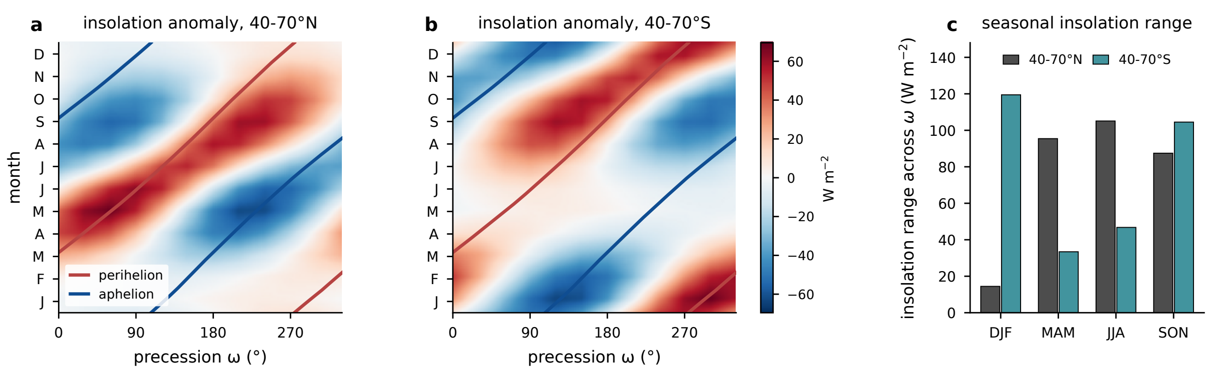

FIG. 1. Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5°. (a) Monthly insolation anomaly relative to the phase mean, averaged over the storm-track band 40–70°N, as a function of calendar month and precession phase ω, with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid. (b) As in (a), but for 40–70°S. (c) Seasonal forcing amplitude in the two bands, the seasonal mean of the monthly maximum minus minimum insolation across the twelve phases. Units: W m^{-2}.

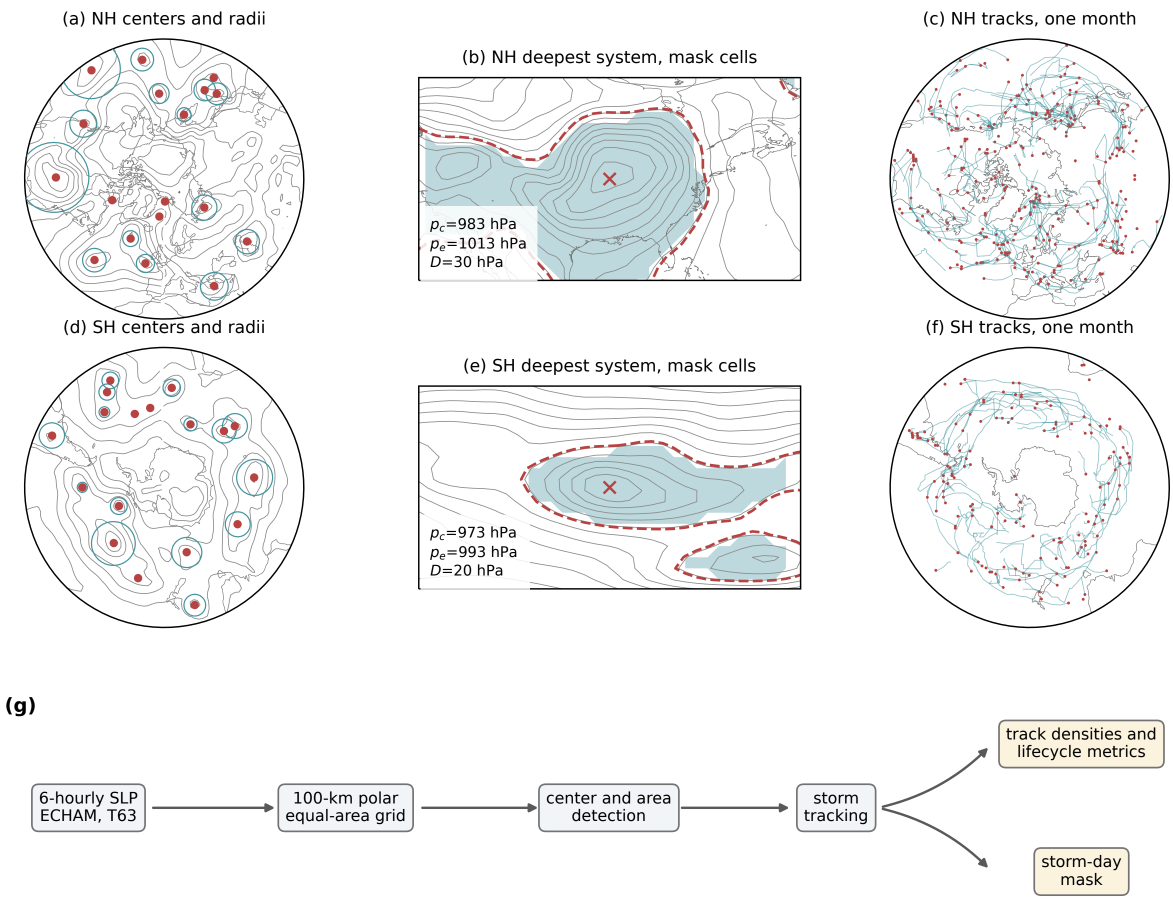

FIG. 2. Cyclone detection and tracking procedure. (a,c) One 6-hourly sea level pressure field of phase p000 (contours, 8-hPa interval) with all detected cyclone centers poleward of 30°N (dots) and their area-equivalent radii $R = \sqrt{A/\pi}$ (circles), 15 January of one model year. (b,e) The deepest system of that field, with the outermost closed isobar p_e (dashed), the cyclone center (cross), the central and edge pressures and depth annotated, and the binary mask cells (shading) defined by SLP < p_e inside the $\sqrt{A}$ search window; contour interval 4 hPa. (c,f) All cyclone tracks of the same month (lines) with genesis points (dots). (d) Data processing workflow.

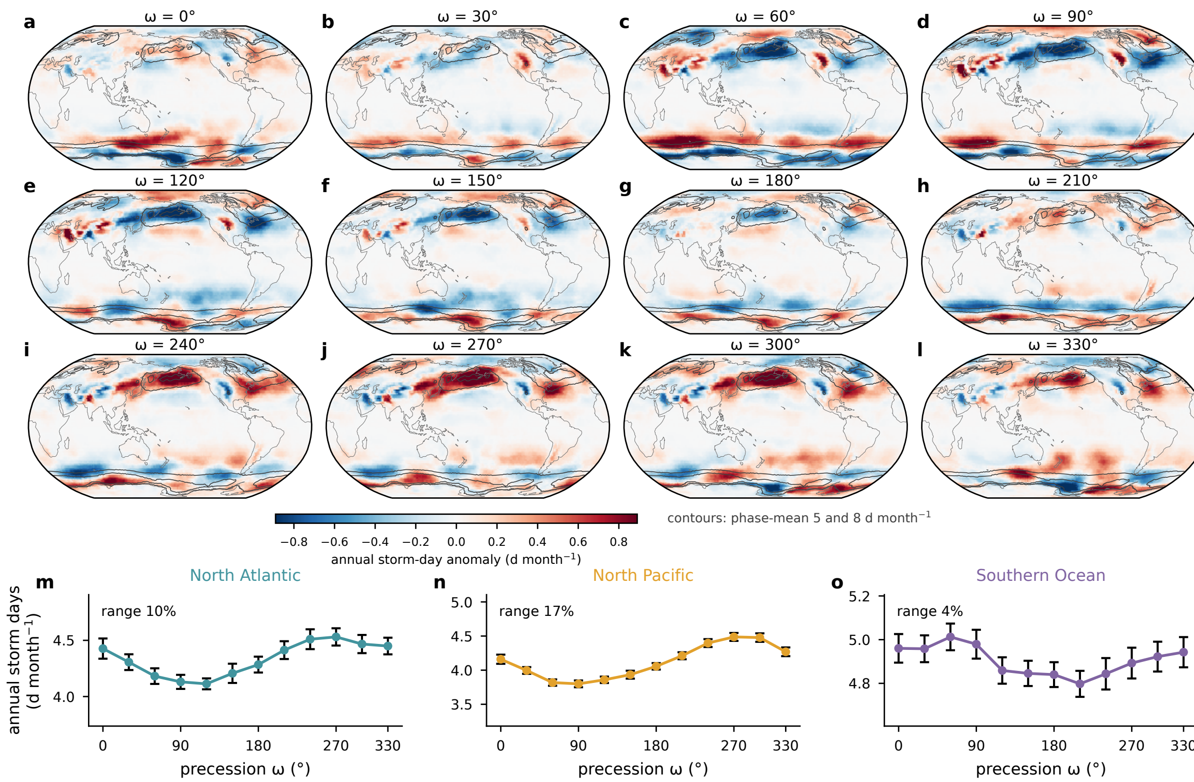

FIG. 3. Annual-mean storm days across the precession cycle. (a)–(l) Annual storm-day anomaly of each phase relative to the twelve-phase mean (shading), with the phase-mean climatology outlined at 5 and 8 d month^{-1} (contours). (m)–(o) Basin-mean annual storm days against ω with interannual standard errors; the across-phase range as a percentage of the phase mean is annotated in each panel.

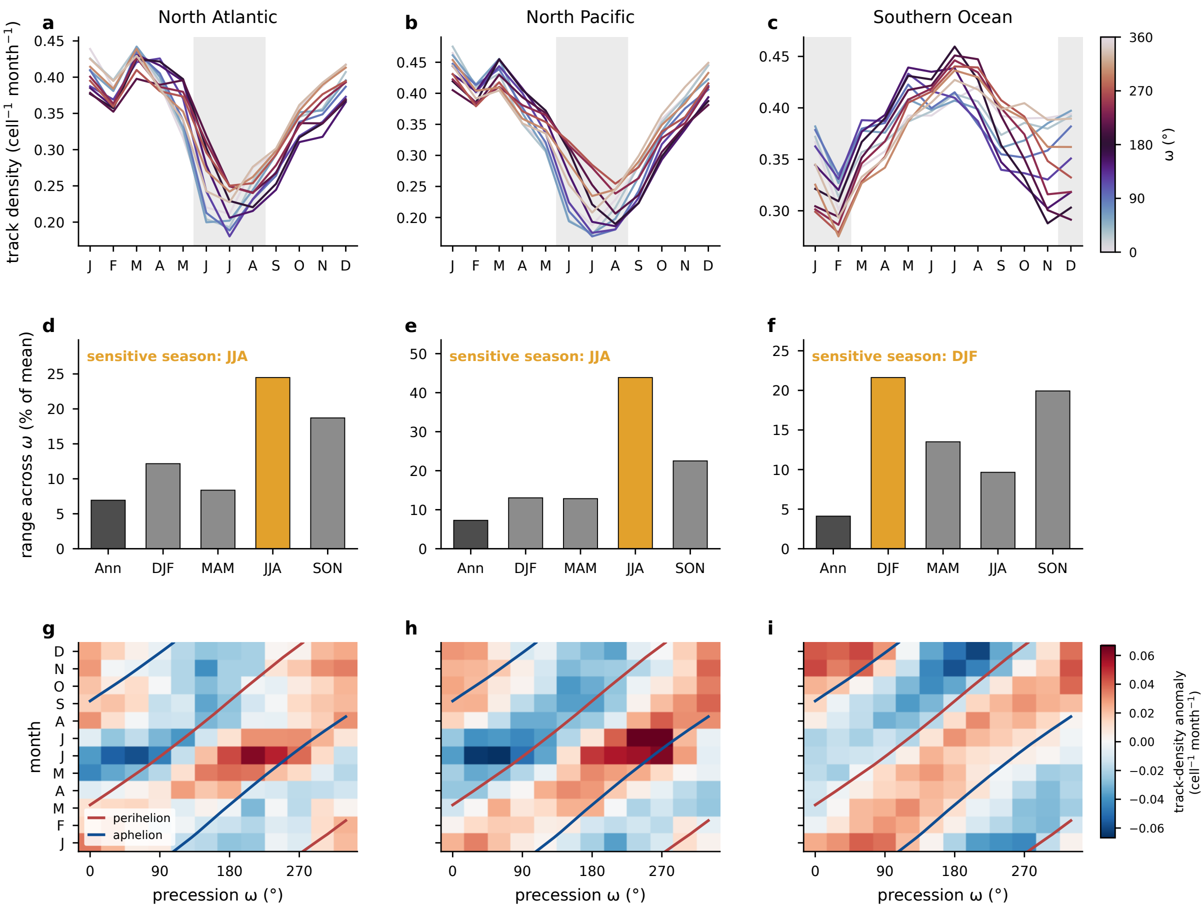

FIG. 4. Seasonal concentration of the precessional storm response, from the Lagrangian tracks. (a)–(c) Basin-mean track density (cell^{-1} month^{-1}) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by ω, with local summer shaded. (d)–(f) Across-phase range of track density (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted; open symbols give the corresponding ranges of system and genesis densities. (g)–(i) Basin-mean track-density anomaly relative to the phase mean as a function of calendar month and precession phase (unsmoothed), with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

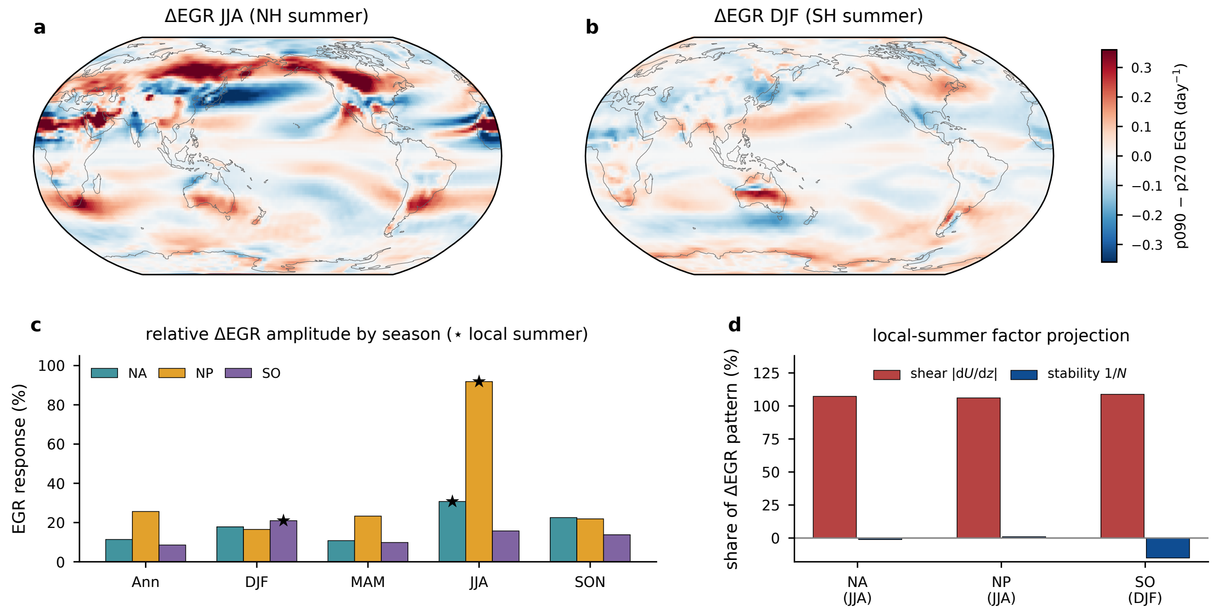

FIG. 5. Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day^{-1}). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

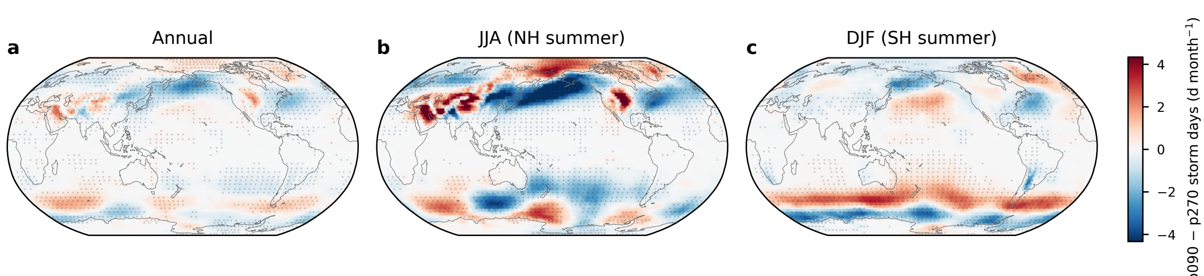

FIG. 6. Difference in storm days (d month^{-1}) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch t test over the 30 simulated years.

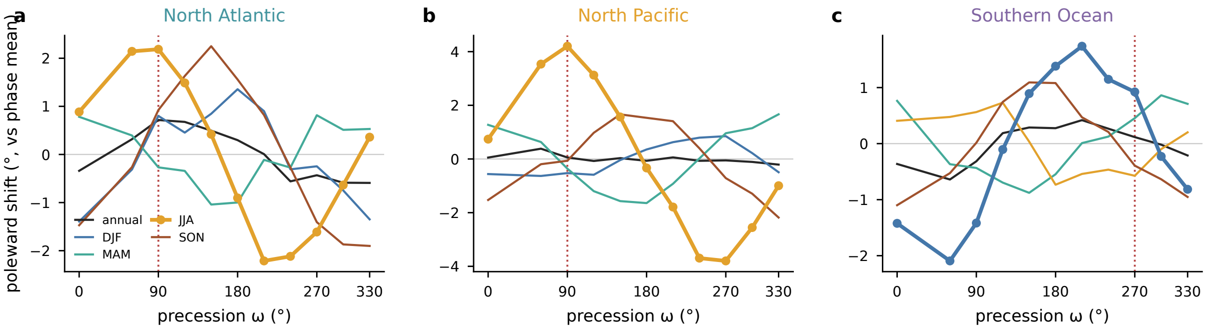

FIG. 7. Storm-track latitude across the precession cycle, the storm-day-weighted mean latitude over each basin sector shown as the poleward anomaly from each season's phase mean, for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean. Lines give the annual mean and the four seasons, with the local summer of each basin drawn heavy with markers. Vertical dotted lines mark the phase with perihelion in the local summer of each basin.

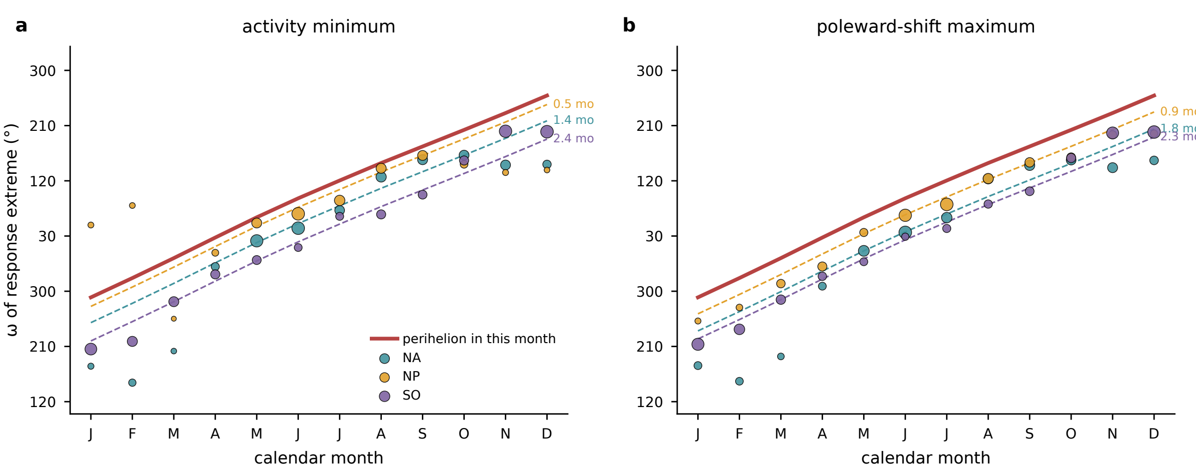

FIG. 8. Phase locking of the storm response to the perihelion calendar. For each calendar month and basin, the phase of the first harmonic across the 12 precession phases of (a) basin-mean storm days, shown as the ω of the activity minimum, and (b) the storm-day-weighted track latitude, shown as the ω of the poleward maximum. Marker size scales with the harmonic amplitude. The red line gives the ω that places perihelion in that calendar month, and dashed lines repeat it delayed by each basin's median lag over the amplitude-strong months (annotated in months).

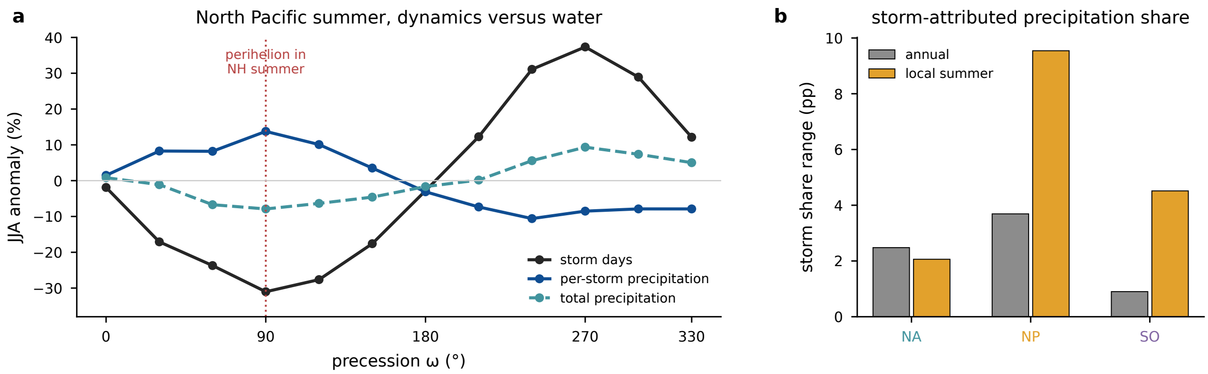

Fig. 9. Storm-attributed precipitation across the precession cycle. (a) North Pacific JJA percent anomalies from the phase mean of basin-mean storm days, per-storm precipitation, and total precipitation; the dotted line marks the phase with perihelion in NH summer. (b) Across-phase range of the storm-attributed share of precipitation, accumulated precipitation inside storm footprints divided by total accumulated precipitation (percentage points), for the annual mean and local summer in each basin.

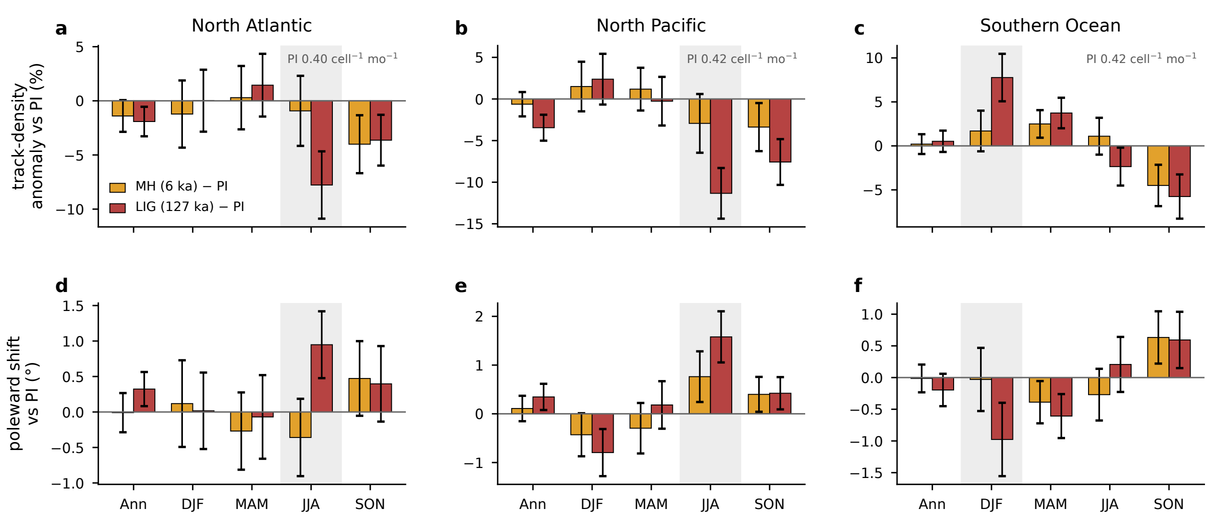

Fig. 10. Seasonal storm response of the real-orbit experiments, tracked with the same pipeline as the main ensemble and referenced to the preindustrial control. (a)–(c) Anomalies of basin-mean track density for the mid-Holocene (gold) and the Last Interglacial (red), as percentages of the preindustrial mean, for the annual mean and the four seasons, with error bars giving twice the standard error of the 30-year difference and the local summer shaded. Track density is normalized per T63 grid cell, and the annotated preindustrial values give the absolute scale in the units of Fig. 4. (d)–(f) The corresponding change in the track-point mean latitude, positive poleward.

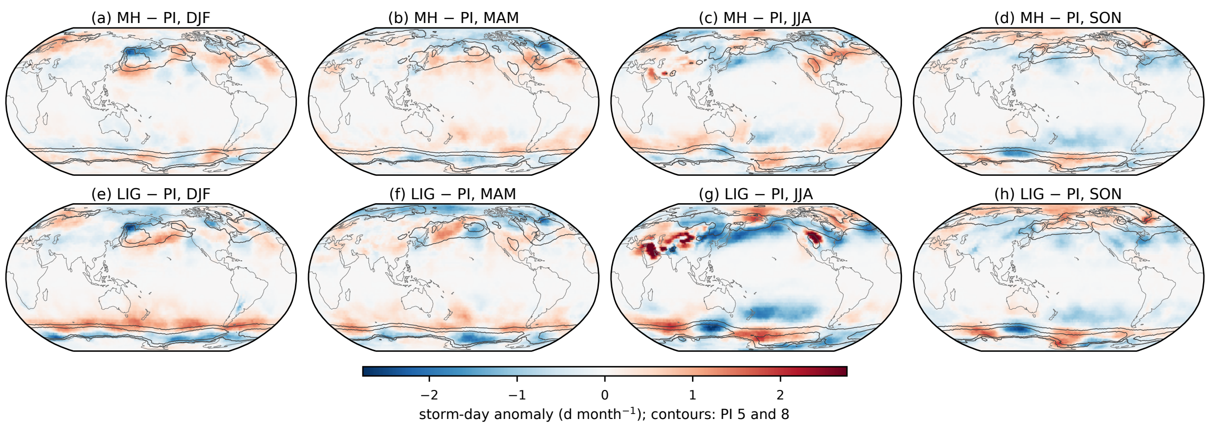

FIG. 11. Storm-day anomalies of the real-orbit experiments relative to the preindustrial control, from the same storm-day product as the main ensemble. (a)–(d) Mid-Holocene minus preindustrial for DJF, MAM, JJA, and SON. (e)–(h) Last Interglacial minus preindustrial. Contours outline the preindustrial climatology at 5 and 8 d month^{-1}. The isolated warm-season maxima over subtropical continental interiors reflect stationary surface heat lows rather than traveling cyclones.

Table 1. Three-basin comparison of the precessional storm response. Ranges are max minus min across the twelve phases as a percentage of the phase mean. Storm and sensitive seasons refer to local winter and local summer. The storm season is diagnosed from cyclogenesis counts, the sensitive season from the seasonal storm-day range. The EGR response is the RMS p090−p270 difference over the basin sector relative to its climatology, and the shear share is the projection of the vertical-shear factor onto the response pattern.

Basin	Storm season	Sensitive season	Storm-day range ann/summer (%)	Summer max/min ω (°)	EGR response summer (%)	Shear share (%)
North Atlantic	DJF	JJA	10 / 32	240 / 90	31	107
North Pacific	DJF	JJA	17 / 68	270 / 90	92	106
Southern Ocean	JJA	DJF	4 / 16	60 / 210	21	109

Acknowledgments. The simulations were performed at the German Climate Computing Center (DKRZ). [TODO funding and grant numbers before submission.]

Data availability statement. The AWI-ESM simulations of the idealized precession ensemble are described in Shi et al. (2025). The cyclone-track database, the gridded storm statistics, and the analysis scripts underlying all figures will be archived and assigned a DOI upon acceptance. [TODO repository DOI.]

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